# Week 4: TA Problems and Solutions

Tashkent University of Technology | Linear Algebra I | Fall 2026

**Matrix rank and matrix inverse**

15 alternative problems with solutions, using different numbers from the lecture. Suggested 80-minute session: recap 5 min; problems 1-5, 15 min; 6-9, 25 min; 10-12, 20 min; a selected challenge and discussion, 15 min.

Ask for reasons as well as arithmetic: which columns are independent, why an inverse exists, and what each row operation accomplishes. Use problems 13-15 selectively as extensions.

All vectors use column notation. Every problem starts on a new PDF page. Statements and solutions are separated by generous blank space. The lecture also has a separate Beamer presentation with alternating problem and solution slides.

## Practice problems

Recall the central equivalence for a square $n \times n$ matrix:

$$A^{-1} \text{ exists} \iff \operatorname{rank} A = n \iff \det A \neq 0.$$

Rank equals the number of pivots. For an inverse, apply row operations to the entire augmented matrix: $[A \mid I] \to [I \mid A^{-1}]$.

Watch for three common errors: counting nonzero rows before reduction; forgetting the right-hand block in a row operation; and failing to reverse factors in the inverse of a product.

### Problem 1: Ranks by inspection - basic

Find the rank of each matrix and give a reason:

$$Z = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}.$$

#### Solution 1

$Z$ has no independent nonzero column, so its rank is 0. In $B$, column 2 is 3 times column 1, and column 1 is nonzero; its rank is 1. The two columns of $D$ point along different coordinate axes, so its rank is 2.

**Problem 2: Independent and dependent columns - basic**

Let $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$. Express column 3 as a combination of columns 1 and 2. Determine the rank and describe the plane containing every output $A\mathbf{x}$.

**Solution 2**

Column 3 equals 3 times column 1 plus column 2. The first two columns are independent because their first two coordinates are $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Therefore the rank is 2. Each column satisfies $z = x + y$, so every output lies in that plane.

**Problem 3: Rank by row reduction - basic**

Reduce $C = \begin{bmatrix} 1 & 3 & 0 & 1 \\ 2 & 7 & 1 & 3 \\ 3 & 10 & 1 & 4 \end{bmatrix}$ to echelon form. State its rank and select independent columns from the original matrix.

**Solution 3**

Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, then $R_3 \leftarrow R_3 - R_2$:

$$C \longrightarrow \begin{bmatrix} 1 & 3 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are two pivots, in columns 1 and 2. The rank is 2, and columns 1 and 2 of the original $C$ are an independent pair.

### Problem 4: Confirm rank with a minor - basic

For $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$, use its determinant and a $2 \times 2$ minor to establish the exact rank. Why is the full determinant alone insufficient?

### Solution 4

The third column is a combination of the first two, so the full determinant is zero and rank is at most 2. The upper-left $2 \times 2$ submatrix is $I_2$, whose determinant is 1. Thus rank is at least 2, and therefore exactly 2. A zero $3 \times 3$ determinant permits ranks 0, 1, or 2.

### Problem 5: Diagonal inverse - basic

Find the inverse of $D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$. State what happens to its rank and inverse if its last diagonal entry is replaced by zero.

### Solution 5

Undo each coordinate scaling:

$$D^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}.$$

The diagonal products are all 1, so $DD^{-1} = I_3$. If the final entry is zero, the rank drops to 2 and the third coordinate is lost; no inverse exists.

### Problem 6: A 2 by 2 inverse - intermediate

Find $A^{-1}$ for $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$ using the determinant formula. Verify your answer by computing $AA^{-1}$.

**Solution 6**

The determinant is $3(1) - (2)(1) = 1$. Therefore

$$A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The nonzero determinant also confirms that the rank is 2.

### Problem 7: Recover an input column - intermediate

Solve $A\mathbf{x} = \mathbf{b}$ using the inverse, where

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}.$$

Check your solution in the original equation.

### Solution 7

Use the inverse from the $2 \times 2$ formula:

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

Substitution gives $A \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$, as required. Because $A$ is invertible, this solution is unique.

### Problem 8: Gauss-Jordan with a row swap - intermediate

Use Gauss-Jordan elimination to invert $H = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}$. Show the augmented matrix after the row swap and at the end.

**Solution 8**

Swap rows 1 and 2, then replace $R_1$ by $R_1 - 3R_2$:

$$\left[\begin{array}{cc|cc} 0 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right].$$

$$\longrightarrow \left[\begin{array}{cc|cc} 1 & 0 & -3 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right].$$

Thus $H^{-1} = \begin{bmatrix} -3 & 1 \\ 1 & 0 \end{bmatrix}$. A zero leading entry required a swap, not a declaration that the matrix was singular.

**Problem 9: Gauss-Jordan for a 3 by 3 matrix - intermediate**

Find the inverse of $U = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ by applying row operations to $[U \mid I_3]$. Verify the upper-right entry of $UU^{-1}$.

**Solution 9**

Use $R_2 \leftarrow R_2 - 3R_3$, then $R_1 \leftarrow R_1 - 3R_2$ with the updated row 2. The right block changes as follows:

$$I_3 \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow U^{-1} = \begin{bmatrix} 1 & -3 & 9 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

The left block becomes $I_3$. The upper-right entry of $UU^{-1}$ is $9 + 3(-3) = 0$, as required.

### Problem 10: Inverse of a product - intermediate

Let $S = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Find $(SD)^{-1}$ without applying the $2 \times 2$ formula directly to $SD$. Check the order of the factors.

**Solution 10**

Reverse the factor order:

$$(SD)^{-1} = D^{-1}S^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{2} \\ 0 & \frac{1}{3} \end{bmatrix}.$$

Indeed, $(SD)(D^{-1}S^{-1}) = SIS^{-1} = I$. The other order gives $S^{-1}D^{-1} = \begin{bmatrix} \frac{1}{2} & -1 \\ 0 & \frac{1}{3} \end{bmatrix}$, a different matrix.

### Problem 11: Prove that an inverse does not exist - intermediate

Let $B = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$. Find a nonzero column $\mathbf{z}$ with $B\mathbf{z} = \mathbf{0}$. Use it to explain why $B$ has no inverse. State its rank.

**Solution 11**

Take $\mathbf{z} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$. Then

$$B\mathbf{z} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

If an inverse existed, multiplication by it would force $\mathbf{z} = B^{-1}\mathbf{0} = \mathbf{0}$, a contradiction. Since column 2 is 3 times the nonzero first column, the rank is 1.

### Problem 12: Rank and information loss in xyz space - intermediate

Let $P = \text{diag}(1, 1, 0)$ and $L = \text{diag}(1, 2, 3)$. Find their ranks. Compute the images of $\mathbf{a} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ under $P$. Which transformation can be reversed?

#### Solution 12

$P$ has rank 2, and

$$P\mathbf{a} = P\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}.$$

It loses the third coordinate, so it cannot be reversed. $L$ has rank 3 and inverse $\text{diag}(1, \frac{1}{2}, \frac{1}{3})$. Its three coordinate scale factors are all nonzero.

### Problem 13: Rank changes with a parameter - challenge

For

$$A(k) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & k-3 & 0 \\ 0 & 0 & k+1 \end{bmatrix},$$

find the rank for every real $k$. For which $k$ does the inverse exist? Find that inverse when it exists.

### Solution 13

The determinant is $(k-3)(k+1)$. The rank is 3 except at $k = 3$ or $k = -1$; at each of those values exactly two independent columns remain, so the rank is 2. For $k \neq 3, -1$,

$$A(k)^{-1} = \begin{bmatrix} 1 & -\frac{1}{k-3} & 0 \\ 0 & \frac{1}{k-3} & 0 \\ 0 & 0 & \frac{1}{k+1} \end{bmatrix}.$$

Multiplication cancels the upper off-diagonal entry and gives $I_3$.

### Problem 14: A one-sided inverse - challenge

Let $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Compute $AB$ and $BA$. Does $B$ qualify as an ordinary two-sided inverse of $A$? State the ranks of $A$ and $B$.

### Solution 14

The products are

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$AB = I_2$, but $BA \neq I_3$. Thus $B$ is a right inverse of $A$, not a two-sided inverse. Both matrices have rank 2; full rank for rectangular matrices does not imply a two-sided inverse.

### Problem 15: Solve a matrix equation - challenge

Find the matrix $X$ satisfying $AXB = C$, where

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 17 \\ 1 & 6 \end{bmatrix}.$$

Show which side you multiply on at each step.

### Solution 15

Multiply on the left by $A^{-1}$, then on the right by $B^{-1}$:

$$X = A^{-1}CB^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 17 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

The order matters because matrix multiplication is not generally commutative. Substitution gives $A \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} B = C$.